Fundamentals of Geographic Information Science (ESS 164 / EARTHSYS 144)

Spring 2026 Syllabus (Boilerplate Draft)

Course Description

"Everything is somewhere, and that somewhere matters."

This course introduces core ideas and methods in Geographic Information Science (GISci), with an emphasis on practical workflows for creating, managing, analyzing, and presenting spatial data. Students work with contemporary GIS tools and real-world datasets to build spatial reasoning and technical fluency.

Primary topics include:

- Geographic data models and spatial thinking
- Coordinate systems and map projections
- Data creation, editing, and field collection
- Tabular operations, SQL, and data integration
- Raster and vector analysis workflows
- Cartographic design and communication
- Introductory remote sensing and web mapping

Learning Objectives

By the end of the course, students should be able to:

1. Use GIS software and cloud geospatial tools in academic and applied settings.
2. Evaluate spatial data quality, coordinate systems, metadata, and fitness for use.
3. Build reproducible workflows for data creation, cleaning, analysis, and visualization.
4. Communicate spatial evidence through clear maps, graphics, and written interpretation.
5. Apply foundational geospatial methods to a student-defined spatial question.

Teaching Team

Instructor

- Stace Maples (maples@stanford.edu)

Teaching Assistants

- Zoie Chang (zochang@stanford.edu)
- Maya Passmore (marykwas@stanford.edu)
- Serena Turner (serena25@stanford.edu)

Course Logistics

Lecture

- Days/Time: Monday and Wednesday, 1:30 PM - 2:50 PM
- Location: Building 380, Room 380Y

Lab Sections

Students should enroll in one lab section. If you have conflicts with the current Lab Section offerings. TA Lab assignments are subject to adjustment.

Section	Day/Time	Location	TA
02	Tuesday, 1:30 PM - 4:20 PM	Y2E2, Room 184	Zoie Chang
03	Thursday, 1:30 PM - 4:20 PM	Y2E2, Room 184	Maya Passmore
04	Friday, 12:30 PM - 3:20 PM	Y2E2, Room 184	Serena Turner

Cross-listed sections (ESS 164 and EARTHSYS 144) share the same meeting times and spaces.

Final Exam (Registrar Scheduled)

- Date: Monday, June 8, 2026
- Time: 3:30 PM - 6:30 PM
- Location: TBD

Office Hours and Contact

- Office hours: Weekly times will be announced in Week 1 and posted on Canvas.
- Format: Office hours will be held in person and by Zoom.
- Quick help: Use the Earthsys144 Slack workspace for short questions.
- Email: For course support, email the instructor and TAs together and include `EARTHSYS 144` in the subject line.

Grading (Boilerplate)

- Weekly Labs and Exercises: 50%

- Midterm Assessment: 20%
- Final Assessment: 20%
- Mini-Project: 10%

Assignments and Assessments

- Lab work is assigned weekly and typically due the following week.
- Midterm assessment: Released Wednesday, May 13, 2026 (8:00 AM), due Monday, May 18, 2026 (11:59 PM).
- Final assessment aligns with university final exam policy and/or assigned take-home format (see announcements).
- Mini-project proposal and final deliverable dates: see Key Dates section.

Important Dates and Deadlines (Spring 2026)

University and Enrollment Deadlines

- March 30 (Mon): First day of instruction
- March 30 (Mon, 5:00 PM): Preliminary Study List deadline
- April 17 (Fri, 5:00 PM): Final Study List deadline (last day to add/drop classes)
- May 11 (Mon, 5:00 PM): Term withdrawal deadline
- May 22 (Fri, 5:00 PM): Change of grading basis deadline
- May 22 (Fri, 5:00 PM): Course withdrawal deadline
- May 25 (Mon): Memorial Day holiday (no classes)
- June 3 (Wed): Last day of classes
- June 5-10 (Fri-Wed): End-quarter examination period

Course and Exam Milestones

- Mini-project proposal due: Friday, April 24, 2026 (11:59 PM)
- Midterm released: Wednesday, May 6, 2026 (12:00 AM)
- Midterm due: Monday, May 11, 2026 (11:59 PM)
- June 8 (Mon, 3:30-6:30 PM): Scheduled final exam slot
- Finals week: Final assessment due (if take-home format is used)
- Mini-project final deliverable due: Sunday, June 7, 2026 (11:59 PM)

Grading System Deadlines

- May 26 (Tue): Grade rosters open for Spring quarter
- June 12 (Fri, 11:59 PM): Grades due for graduating students
- June 16 (Tue, 11:59 PM): Grades due for non-graduating students

Tentative 10-Week Course Schedule (Spring 2026)

Dates	Week	Topic	Lecture Focus	Suggested Reading
Mar 30 & Apr 1	1	Intro to GISci and Cartographic Design	Spatial data models, map purpose, design basics	Bolstad Ch. 1-2
Apr 6 & Apr 8	2	Coordinates, Projections, and Data Creation	Geodesy, projection choice, digitizing workflows; Apr 8 guest lecture: Ben Gitai (Atlas of Paris Landscapes, SF Chronomapping)	Bolstad Ch. 3-4, 7
Apr 13 & Apr 15	3	Tables, SQL, and Data Structure	Relational tables, joins, filters, query logic	Bolstad Ch. 5, 8
Apr 20 & Apr 22	4	Vector Analysis Fundamentals	Buffer, overlay, proximity, areal workflows; Apr 22 guest lecture: TBD	Bolstad Ch. 9, 12
Apr 27 & Apr 29	5	Applied GIS Workflows	Case studies, domain applications, project design	Supplemental readings
May 4 & May 6	6	Raster Analysis and Terrain	Raster algebra, sampling, interpolation concepts	Bolstad Ch. 10-12
May 11 & May 13	7	Remote Sensing and Web Mapping	Sensor basics, imagery interpretation, web maps	Bolstad Ch. 6
May 18 & May 20	8	Storytelling and Communication	Story maps, narrative structure, visual argument	Supplemental readings
May 27 (Mon holiday week)	9	Analytic Models and Synthesis	Model logic, assumptions, and interpretation	Bolstad Ch. 15
Jun 1 & Jun 3	10	Final Integration and Review	Advanced examples, review, project support; Jun 1 guest lecture: TBD	Instructor-provided resources

Software and Materials

- Required: QGIS (latest release)
- Used in course: Google Earth Engine, ArcGIS Online, and selected open-source tools
- Hardware: Laptop computer with reliable internet access

Access and Accommodations

Students needing academic accommodations should register with the Office of Accessible Education (OAE) and share accommodation letters as early as possible.

Notes

- This is a boilerplate syllabus draft and is subject to revision.
- Course policies, assignment details, and exact due dates may be updated in class and on Canvas.